

Exam Date & Time: 14-Jun-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VI Semester Makeup Examination, June -2025

Artificial Intelligence [CSE-4053]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

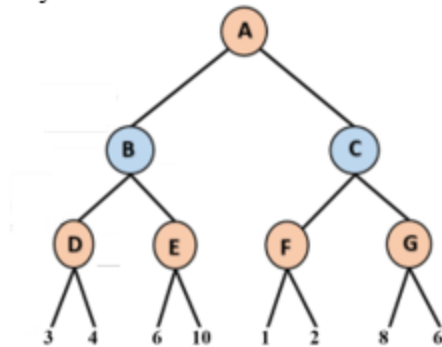
- 1) Differentiate between simple reflex agents and model-based agents. Highlight their architecture, decision-making approach, and real-world applicability with suitable examples. (5)
 - A)
 - B)
- 2) Compare and contrast between expert task and mundane task in the domain of artificial intelligence. Apply these themes and present suitable examples. (5)
 - A)
 - B)
- 3) Design a hybrid intelligent agent architecture combining simple reflex and model-based reflex agents for a Traffic Management System. Explain the types of sensors and actions involved and how each agent type responds to real-time traffic scenarios. (5)
 - A)
 - B)
- 4) Explain how Depth-First Search and Breadth-First Search behave when applied to large state spaces. Evaluate their suitability for real-time and memory-constrained AI systems. (5)
 - A)
 - B)
- 5) Consider the following graph with nodes A to F, edge costs, and heuristic values:
 - A)

Edge	Cost	Heuristic Values
A – B	2	$h(A) = 10$
A – C	4	$h(B) = 8$
B – D	7	$h(C) = 5$
C – D	1	$h(D) = 3$
C – E	3	$h(E) = 6$
D – F	1	$h(F) = 0$
E – F	5	$h(G) = 4$
B – G	4	$h(H) = 2$
E – G	2	
G – H	3	
H – F	2	

Apply A* algorithm to find the optimal path from A to F."

- 6)
 - A)
 - B)

Consider the following game tree, where each leaf node is labeled and contains a utility value. Apply alpha-beta pruning to this tree. Clearly show the alpha and beta updates at each node. Mark pruned branches, if any and identify the optimal value returned by node A.



- 4) Construct a 2D matrix representation of the world using the following placement:
- Pit at (1,2), (3,3); Wumpus at (2,3); Gold at (4,4) (5)
- A) You have to mark cells affected by breeze or stench using adjacency rules. Show how an agent would use this grid to infer safe paths.
- B) Generate equivalent FOPL statements for the followings:
- i) All students submit assignments on time
 - ii) Some teachers use digital boards (5)
 - iii) No vehicles are allowed inside the campus
 - iv) Every robot follows instructions
- 5) Examine and analyze the structural and expressive differences between propositional logic and predicate logic. Compare their capabilities, limitations, and suitability for representing real-world AI knowledge with examples. (5)
- A)
- B) An AI system for disease detection uses Bayesian reasoning. Given the following probabilities:
- $P(\text{Cough}) = 0.25$
- $P(\text{Tuberculosis}) = 0.05$
- $P(\text{Cough} \mid \text{Tuberculosis}) = 0.85$ (5)
- Task:
- Use Bayes' Theorem to compute the probability that a person has tuberculosis given that they have a cough, i.e., $P(\text{Tuberculosis} \mid \text{Cough})$. Clearly explain each step in your calculation.

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